

ATPL EASA STUDY GUIDE | 061 - GENERAL NAVIGATION

Chapter 10: In-Flight Navigation & The 1-in-60 Rule

Syllabus Source: CAE Oxford (pp. 365-408) | Pass Target: $\geq 90\%$ AviationExam

1. THE GEOMETRY OF THE 1-IN-60 RULE

The 1-in-60 rule is an empirical trigonometric approximation: An angular displacement of 1 DEGREE produces an off-track distance of exactly 1 NAUTICAL MILE after flying 60 NAUTICAL MILES ($\tan 1^\circ \sim 1/60$):

- Track Error Angle (TE): The angular divergence between the planned track and the actual track flown so far: $\text{Track Error (degrees)} = (\text{Distance Off Track} / \text{Distance Gone}) \times 60$.
- Closing Angle (CA): The additional angle required to intercept the next planned waypoint: $\text{Closing Angle (degrees)} = (\text{Distance Off Track} / \text{Distance to Go}) \times 60$.
- Heading Alteration (To fly DIRECT to next waypoint): $\text{Alteration of Heading} = \text{Track Error} + \text{Closing Angle}$ (Change Heading = TE + CA). (If you only correct by TE, you will fly parallel to the track but never regain it!).

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): The 1-in-60 In-Flight Course Correction

Problem: Flight leg is 120 NM from A to B. After 40 NM, aircraft is 4 NM RIGHT of track. What total heading correction will intercept destination B directly? Step 1: Dist flown = 40 NM. Dist to go = $120 - 40 = 80$ NM. Step 2: Track Error Angle (TE) = $(\text{Dist Off} \times 60) / \text{Dist Flown} = (4 \times 60) / 40 = 6^\circ$. Step 3: Closing Angle (CA) = $(\text{Dist Off} \times 60) / \text{Dist to Go} = (4 \times 60) / 80 = 3^\circ$. Step 4: Total Alteration = $\text{TE} + \text{CA} = 6^\circ + 3^\circ = 9^\circ$. Step 5: Turn 9° LEFT (subtract 9° from present heading) to fly straight to destination B!

2. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Correcting Heading In-Flight using 1-in-60 Rule

SCENARIO (Classic ATPL In-Flight Correction Drill): A flight leg from Waypoint A to Waypoint B has a total planned distance of 120 NM on Planned Track 090° True: - After flying 40 NM from Waypoint A, the pilot fixes position and discovers the aircraft is 4 NM to the RIGHT of track - Distance remaining to Waypoint B = 120 NM - 40 NM = 80 NM QUESTION: Calculate (1) Track Error, (2) Closing Angle, and (3) The new heading alteration to fly directly to Waypoint B. SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate Track Error (TE): - Formula: $TE = (Distance\ Off / Distance\ Gone) \times 60$ - Distance Off = 4 NM - Distance Gone = 40 NM - $TE = (4 / 40) \times 60 = 0.10 \times 60 = 6^\circ$ Track Error (drifted 6° to the Right)! Step 2: Calculate Closing Angle (CA): - Formula: $CA = (Distance\ Off / Distance\ to\ Go) \times 60$ - Distance Off = 4 NM - Distance to Go = 80 NM - $CA = (4 / 80) \times 60 = 0.05 \times 60 = 3^\circ$ Closing Angle! Step 3: Calculate Total Heading Alteration to fly directly to Waypoint B: - Total Alteration = Track Error (TE) + Closing Angle (CA) - Total Alteration = $6^\circ + 3^\circ = 9^\circ$ to the LEFT! Step 4: Determine the New Heading: - Aircraft is currently drifted to the RIGHT of track. - The pilot must turn 9° to the LEFT to eliminate track error and intercept Waypoint B! - If old heading was 090°, New Heading = $090^\circ - 9^\circ = 081^\circ$ True! FINAL ANSWER: Track Error = 6°; Closing Angle = 3°; Alter heading 9° LEFT to 081° True.

[!] EXAM TRAP (AVIATIONEXAM) - The TE Only vs TE + CA Trap

- If you alter heading by TRACK ERROR ONLY (6° Left): - You will fly PARALLEL to the original track, remaining 4 NM off track forever! - To REGAIN THE DESTINATION, you MUST ADD the Closing Angle (TE + CA = 9°)! Examiners frequently test this exact distinction.

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: After flying 30 NM along a planned route of 90 NM, an aircraft is 3 NM left of track. What is the TRACK ERROR?

- [A]: 3°
- [B]: 6° ($TE = (3\text{ NM} / 30\text{ NM}) \times 60 = 0.10 \times 60 = 6^\circ$)
- [C]: 9°
- [D]: 1°

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Track error = $(Distance\ Off / Distance\ Gone) \times 60 = (3 / 30) \times 60 = 6^\circ$ left of track.

Q2: In the scenario above (3 NM off track, 60 NM to go), what is the CLOSING ANGLE (CA)?

- [A]: 6°
- [B]: 3° ($CA = (3\text{ NM} / 60\text{ NM}) \times 60 = 3^\circ$)
- [C]: 9°
- [D]: 1.5°

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Closing angle = $(Distance\ Off / Distance\ to\ Go) \times 60 = (3 / 60) \times 60 = 3^\circ$.

Q3: Continuing from above, by how many degrees must the pilot alter heading to fly DIRECTLY to the destination

- [A]: 6° Right
- [B]: 9° RIGHT (Alteration = TE + CA = 6° + 3° = 9° Right)
- [C]: 3° Right
- [D]: 12° Right

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. To intercept destination, turn into the error by the sum of TE and CA:
6° + 3° = 9° to the Right.